

# Krystal Luo

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## EDUCATION

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**Carnegie Mellon University**

MS in Biomedical Engineering

August 2025 - December 2026

Pittsburgh, PA

**University of North Carolina at Chapel Hill**

BS in Biomedical Engineering, **Highest Honor**

BA in Biological and Medical Physics, **Highest Honor**

Minored in Mathematics

August 2021 - May 2025

Chapel Hill, NC

**Dean's List**

## PUBLICATIONS

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- **W. Luo**, Y. Hu, A. F. Stadler, A. J. Moretti, D. A. Tyndall, C. R. Inscoe, Y. Z. Lee, J. Lu, et al. "Feasibility of multisource CBCT for improving the predictability of dental implant primary stability compared to conventional CBCT," *Scientific Reports*, 2026.
- Y. Wu, S. Xing, D. Yang, Y. Liu, C. Ding, Z. Li, Q. Jiao, A. Zhang, Z. Guo, S. Liu, **W. Luo**, et al. "Soft giant magnetoimpedance electronics enable contact-free human-machine interactions," *Proceedings of the National Academy of Sciences* 123 (4), 2026.
- Y. Wu, J. Wang, G. Yuan, Y. Chen, K. Liang, D. Yang, Y. Liu, **W. Luo**, S. Xing, et al. "Symmetry engineering in a 2D transition metal enables reconfigurable P-and N-Type FETs," *Nano Letters* 25 (5), 1994-2001, 2025.
- L. Zhang, S. Xing, H. Yin, H. Weisbecker, H. T. Tran, Z. Guo, T. Han, Y. Wang, **W. Luo**, et al. "Skin-inspired, sensory robots for electronic implants," *Nature Communications* 15 (1), 4777, 2024.

## RESEARCH EXPERIENCE

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**Research Assistant**, LOGIC Lab, Carnegie Mellon University

Advisor: Dr. Saurabh Vyas

August 2025 – Present

### Ultra-large-scale brain-wide electrophysiology

- Developed MRI- and CT-guided planning workflows in Brainsight for large-scale Neuropixels-NHP electrophysiology experiments targeting flexible behavior and logical reasoning regions.
- Designed custom mechanical apparatuses and burr-hole guidance tools for precise skull drilling, probe insertion trajectory alignment and stabilization.

### Automated In-Cage Training and Reward Delivery Systems

- Engineered wireless reward delivery systems and automated cage-mounted tablet interfaces utilizing custom PCBs, Bluetooth microcontrollers, and pump-control circuitry.

**Research Assistant**, Bai Lab, University of North Carolina at Chapel Hill

Advisor: Dr. Wubin Bai

August 2022 – May 2025

### MST-DCBS: Multi-Scale Transformer Framework for Spike Sorting and Decoding

- Developed a multi-scale transformer framework utilizing attention blocks and self-supervised learning for extracellular spike sorting and neural decoding.
- Benchmarked sorting accuracy, decoding performance, and computational efficiency on real and simulated high-dimensional datasets for honors thesis research.

### Microelectrode Array Platform for Brain Organoid Recordings

- Fabricated microelectrode arrays via photolithography for brain organoid recordings, validating electrode functionality and stability through impedance testing.
- Processed extracellular signals with adaptive filtering and spike clustering, visualizing neural dynamics via heatmaps, raster plots, and spike-rate analyses.

### Hydrogel-Based Soft Robotics for Drug Delivery and Minimally Invasive Manipulation

- Synthesized thermoresponsive PNIPAM hydrogels and prototyped soft robotic actuators for controlled drug delivery.
- Optimized device geometry and mechanical response through iterative experimental testing and 3D simulations of deformation behavior.

**Research Assistant**, Applied Nanotechnology Lab, University of North Carolina at Chapel Hill

*Advisor: Dr. Otto Zhou*

*January 2024 – May 2025*

#### **Multisource CBCT Imaging for Dental Implant Primary Stability**

- Operated multisource CBCT imaging systems and performed routine calibrations to acquire high-resolution datasets for predicting dental implant primary stability.
- Segmented complex maxillofacial structures using 3D Slicer and extracted quantitative imaging features for structural and stability assessment workflows.

**Research Assistant**, State Key Laboratory of Bioelectronics, Southeast University

*Advisor: Dr. Nongyue He*

*May 2023 – August 2023*

#### **In Vivo Tumor Growth Study and Cell Culture**

- Maintained mammalian cell cultures using sterile techniques and conducted in vivo tumor cell injections in mice under established ethical guidelines.
- Monitored longitudinal tumor growth and health status, maintaining comprehensive laboratory records for experimental assessment.

**Research Assistant**, University of California, Los Angeles

*Advisor: Dr. Dejan Marković*

*June 2022 – August 2022*

#### **Brain-Computer Interface and Flexible Neural Interface Research**

- Investigated flexible, skin-adherent bioelectronic designs and signal acquisition strategies for wearable neural interfaces.
- Reviewed data processing pipelines and signal acquisition methods to enhance continuous brain-computer interface (BCI) reliability.

### **TEACHING EXPERIENCE**

**Teaching Assistant, BMME 301: Human Physiology – Electrical Analysis** August 2023 – May 2025

*Lampe Joint Department of Biomedical Engineering, UNC-Chapel Hill*

- Led laboratory sections on physiological signal acquisition (EKG, EEG, action potentials) and biological measurement techniques.
- Mentored students in experimental design, data interpretation, and technical reporting for hands-on physiological studies.

**Teaching Assistant, BMME 207: Biomedical Electronics**

August 2023 – May 2025

*Lampe Joint Department of Biomedical Engineering, UNC-Chapel Hill*

- Guided students through building, testing, and troubleshooting biomedical electronic circuits and instrumentation.
- Delivered technical lectures bridging theoretical circuit concepts with practical biomedical applications and debugging strategies.

### **AWARDS AND HONORS**

- **BME Department Head's Fellowship**, Carnegie Mellon University March 2025
- **Silver**, Kaggle LLMs – You Can't Please Them All; ranked **33/1,692** March 2025
- **Robert N. Shelton Award for Outstanding Research**, UNC-CH Physics and Astronomy February 2025
- **Bronze**, NeurIPS 2024 Ariel Data Challenge; ranked **59/1,151** October 2024
- **Silver**, RSNA 2024 Lumbar Spine Degenerative Classification Challenge; ranked **49/1,874** October 2024
- **Abrams Scholarship**, Lampe Joint Department of Biomedical Engineering, UNC-CH September 2024